EXPERIMENT - 20

20. Implement the Dilation technique as a Morphological operation to dilate the foreground regions based on Open CV.

Aim

To implement the Dilation technique as a morphological operation using OpenCV in Python to increase the size of foreground regions in an image.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Resize the image to fit the screen.
4. Convert the image into grayscale.
5. Create a kernel using np.ones().
6. Apply dilation using cv2.dilate().
7. Display the original image and dilated output image.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

if img is None:
    print("Image not found")
else:
    img = cv2.resize(img, (500, 350))
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kernel = np.ones((5, 5), np.uint8)

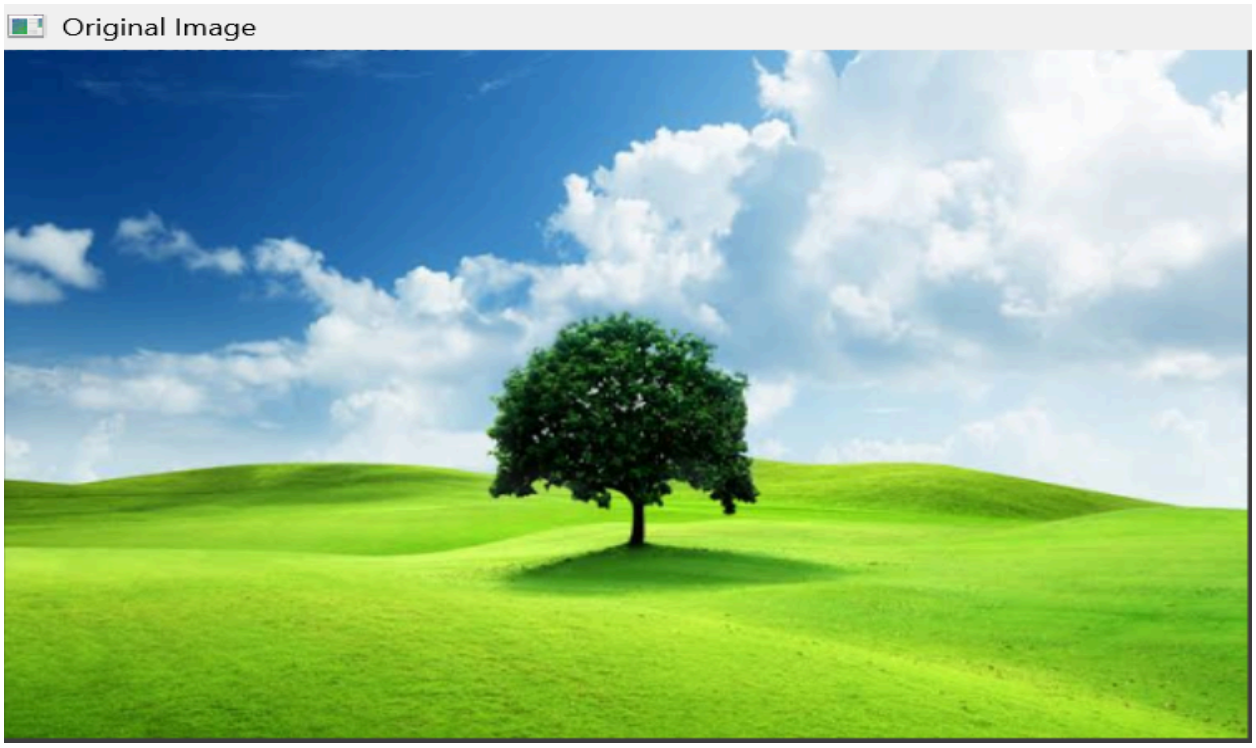
    dilation = cv2.dilate(gray, kernel, iterations=1)

    cv2.imshow("Original Image", img)
    cv2.imshow("Dilation Output", dilation)

    cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input Image



Output Image



Result

The Dilation Morphological Operation was successfully implemented using OpenCV in Python to increase the size of foreground regions.